

A Hybrid LSMC–PDE Method for Bermudan Option Pricing under the Gatheral Double Mean-Reverting Model

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Abstract

We study Bermudan option pricing under the Gatheral Double Mean-Reverting (GDMR) stochastic volatility model. The model features a variance process together with a stochastic long-run mean variance process and allows Constant Elasticity of Variance (CEV)-type exponents in the diffusion coefficients. This model is attractive since it provides a flexible specification for volatility dynamics. However, the pricing of early-exercise derivatives under the GDMR model remains largely unexplored in the literature. To address this challenge, we adapt a Hybrid Least-Squares Monte Carlo–Partial Differential Equation (LSMC–PDE) framework to the GDMR model and provide a detailed model-specific implementation. Conditioning on simulated variance paths, the pricing problem reduces to a one-dimensional problem in the asset price, which is solved by a Fourier-based approach, while the remaining dependence on the variance variables is approximated by least-squares regression. Our numerical experiments demonstrate that the Hybrid LSMC–PDE approach yields accurate pricing estimates and often lower pricing errors than plain LSMC, particularly for low and moderate numbers of simulation paths, showing the benefit of using the model structure in early-exercise option pricing.

Keywords: Bermudan option pricing, stochastic volatility, Hybrid LSMC–PDE, least-squares Monte Carlo, Gatheral model

1 Introduction

Bermudan option pricing under stochastic volatility requires repeated approximation of continuation values defined as conditional expectations of discounted future pay-offs. Deterministic Partial Differential Equation (PDE) methods handle this recursion well in low-dimensional Markovian models, but their cost grows quickly when several volatility factors are present. Under a multi-factor stochastic volatility model, the resulting three or higher dimensional pricing problem becomes increasingly computationally demanding.

Regression-based Monte Carlo approaches for pricing American-style options were developed by Tsitsiklis and Van Roy [1] and later popularized through the Least-Squares Monte Carlo (LSMC) method of Longstaff and Schwartz [2], which is now widely used for pricing options with early exercise features. Clément, Lamberton, and Protter [3] analyzed the convergence of the least-squares regression method for American option pricing. These methods are flexible, but in stochastic-volatility models the regression approximation of the continuation value function must capture the dependence on both asset and volatility variables.

Mixed Monte Carlo–PDE methods, which combine simulation with conditional PDE calculations, have shown to be promising for option pricing problems under stochastic-volatility models; see, for example, Loeper and Pironneau [4], Lipp, Loeper, and Pironneau [5], and Cozma and Reisinger [6]. More recently, Farahany, Jackson, and Jaimungal [7] developed a Hybrid LSMC–PDE method for pricing Bermudan options under stochastic volatility, which forms the methodological basis of the present work. Their algorithm simulates the volatility process, solves conditional asset-direction PDE problems along simulated volatility paths, and then regresses the resulting conditional values over the volatility state. They also prove almost-sure convergence for a class of stochastic-volatility models.

We study Bermudan option pricing under the Gatheral-type double mean-reverting stochastic volatility (GDMR) model, introduced in Gatheral [8], where successful simultaneous calibration to both SPX and VIX option markets was demonstrated. Under this model, the instantaneous variance v mean-reverts to a stochastic variance level v' , while v' mean-reverts to a constant long-run level. This double mean-reverting structure is motivated by the volatility-surface literature (see, e.g., Heston [9], Cox, Ingersoll, and Ross [10], and Gatheral [11]).

Despite its appealing modelling properties, the GDMR model does not admit closed-form pricing formulas, even for European options. As a result, calibration in Gatheral [8] was performed using Euler simulation, while Bayer, Gatheral, and Karlsmark [12] later employed the higher-order Ninomiya–Victoir scheme to improve computational efficiency. Moreover, there has been limited research on Bermudan and American option pricing under this framework. To the best of our knowledge, Haida et al. [13] is among the few studies addressing this problem, proposing a simulation scheme together with a full Least-Squares Monte Carlo (LSMC) method for American option pricing under a double-Heston-type parameterization of the GDMR model. In this work, we aim to fill this gap by developing a Hybrid LSMC–PDE method, adapted from the more general framework of Farahany, Jackson, and Jaimungal [7]. This approach is motivated by the computational challenges of applying a

full PDE method in the presence of three state variables, while avoiding the need for a full regression approximation of the continuation value as in standard LSMC. Although our discussion focuses on Bermudan option pricing, the framework naturally extends to early-exercise derivatives more broadly through Bermudan approximations of American options.

We note two advantages of this hybrid method relative to the plain LSMC method. First, rather than approximating the entire continuation value through a parametric regression approximation, the Hybrid LSMC–PDE method performs regression only partially, leaving the stock-price dependence to be resolved through a conditional pricing PDE. Second, the conditional PDE step naturally produces option values on a grid of stock prices, corresponding to multiple moneyness levels for a fixed strike. This is advantageous for calibration, where option prices across different moneyness values and maturities must be computed repeatedly.

The present paper focuses on adapting the Hybrid LSMC–PDE framework to the GDMR model, motivated by the considerations discussed above. This specialization is nontrivial for the Gatheral double mean-reverting model, because the model has two correlated variance Brownian drivers and Constant Elasticity of Variance (CEV)–type variance coefficients. Our contribution is thus a model-specific formulation and implementation of an existing Hybrid LSMC–PDE framework for the multifactor GDMR setting. First, we derive the Brownian projection and the conditional one-step Gaussian representation needed to apply the existing Hybrid LSMC–PDE framework to the GDMR model. The key model-specific quantities are the correlated Brownian shift Z_n , the integrated drift I_n , the integrated residual variance B_n , and the terminal volatility state.

From a computational perspective, we provide a detailed description of the FFT-based numerical implementation and numerical experimental results which may serve as benchmark values for future studies. For the numerical experiments, we use a fixed parameter set based on a calibrated double mean-reverting specification from Bayer, Gatheral and Karlsmark [12]. We compare plain LSMC estimator and the Hybrid LSMC–PDE estimator by varying both the Euler step count and the Monte Carlo path budget. Since no closed-form Bermudan option price is available under the GDMR model, the comparisons are made against large-simulation plain LSMC reference prices. Our numerical results indicate that the Hybrid LSMC–PDE method yields accurate pricing estimates and often lower pricing errors than plain LSMC method, particularly for low and moderate numbers of simulation paths.

In addition, we also include a short well-posedness result for the variance subsystem of the GDMR dynamics. This result is not the main contribution of the paper, it is included to make explicit the closed nonnegative variance domain used in the Bermudan recursion and in the conditional construction.

The remainder of the paper is organized as follows. Section 2 discusses the GDMR model and gives the well-posedness result. Section 3 gives the theoretical framework for Bermudan option pricing using the Hybrid LSMC–PDE method and, in particular, derives the conditional Hybrid LSMC–PDE representation. Section 4 describes the numerical implementation. Section 5 reports the experiments. Section 6 concludes.

2 Gatheral double mean-reverting model

This section introduces the GDMR model. In this model, the equations for the variance factors do not involve the asset price, while the asset equation depends on the variance factors. This one-way coupling is used in Section 3 to obtain the conditional one-dimensional representation.

2.1 GDMR dynamics

Fix a finite horizon $T > 0$. Let

$$(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \in [0, T]}, \mathbb{Q})$$

be a filtered probability space satisfying the usual conditions. Throughout the paper, option pricing is performed under a risk-neutral (equivalent martingale) measure $\mathbb{Q}$ under which the discounted stock price process is assumed to be a true martingale. We assume a deterministic money-market account $B_t = e^{rt}$ where r is the constant risk-free interest rate. Unless otherwise specified, all expectations are taken with respect to the probability measure $\mathbb{Q}$. Furthermore, the notation $\mathbb{E}_\xi[\cdot]$ denotes expectation with respect to the probability distribution of the random variable ξ .

The probability space supports a three-dimensional correlated $(\mathcal{F}_t)$ -Brownian motion

$$W = (W^{(1)}, W^{(2)}, W^{(3)})$$

with constant quadratic covariations

$$\langle W^{(i)}, W^{(j)} \rangle_t = \rho_{ij} t, \quad 1 \leq i, j \leq 3.$$

The matrix $\rho = (\rho_{ij})_{1 \leq i, j \leq 3}$ is symmetric positive semidefinite and satisfies $\rho_{ii} = 1$. We assume $|\rho_{23}| < 1$. This ensures that the covariance matrix of the two variance Brownian drivers, $W^{(2)}$ and $W^{(3)}$, is nonsingular. The assumption is needed in Section 3 where $W^{(1)}$ is decomposed into its projection onto the span of $(W^{(2)}, W^{(3)})$ and an orthogonal residual Brownian component.

We consider the GDMR dynamics of Gatheral [8], augmented with a risk-free drift term in the stock-price dynamics:

$$\begin{aligned} dS_t &= r S_t dt + S_t \sqrt{v_t} dW_t^{(1)}, \\ dv_t &= \kappa_1 (v'_t - v_t) dt + \xi_1 v_t^{\delta_1} dW_t^{(2)}, \\ dv'_t &= \kappa_2 (\theta - v'_t) dt + \xi_2 (v'_t)^{\delta_2} dW_t^{(3)}, \end{aligned} \tag{1}$$

where $r \geq 0$, $\kappa_1, \kappa_2, \theta, \xi_1, \xi_2 > 0$, and $\delta_1, \delta_2 \in [1/2, 1]$. The factor v is the instantaneous variance and mean-reverts to the stochastic level v' , while v' mean-reverts to the long-run level θ . The exponents δ_1 and δ_2 , referred to as *CEV exponents*, replace the square-root volatility-of-volatility coefficients by CEV-type powers; the square-root double mean-reverting model is recovered when $\delta_1 = \delta_2 = 1/2$.

We further restrict the leverage correlations to satisfy $\rho_{12}, \rho_{13} \in [-1, 0]$. In particular, negative stock-volatility correlations are known to mitigate potential moment explosion effects and support stable martingale behavior in stochastic volatility models as pointed out in Andersen and Piterbarg [18]. This assumption is also consistent with market stylized facts on stock-volatility correlations.

2.2 Well-posedness and nonnegativity

The purpose of this subsection is to state the state-space property of the variance dynamics used later in the pricing construction. Since the variance factors are simulated first and then used in the conditional Hybrid LSMC–PDE step, the factors v and v' must be well defined and nonnegative on the exercise horizon.

The only delicate point is the boundary at zero. When $\delta_i < 1$, the CEV coefficients are not Lipschitz at the boundary, so the usual globally Lipschitz SDE argument does not apply directly. This is why we state the well-posedness and nonnegativity result explicitly.

Mishura, Pilipenko, and Ralchenko [17] also study the Gatheral double stochastic-volatility system. In their notation, the variance factors are X and Y , corresponding to v and v' in (1), with $a_1 = b_1 = \kappa_1$, $a_2 = \kappa_2\theta$, $b_2 = \kappa_2$, $\sigma_i = \xi_i$, and $\alpha_i = \delta_i$. They assume strictly positive deterministic initial values and prove strong existence, uniqueness, nonnegativity, and the strong Markov property. Their paper then studies a different issue: whether the volatility factors can remain close to zero, and how a Skorokhod-reflected modification of the internal factor can prevent this behaviour.

We keep the risk-neutral GDMR dynamics in Eq. (1), and we use the result only to justify the closed nonnegative variance domain needed for the Bermudan dynamic programming and for the conditional construction in Section 3. Thus, the theorem is stated in the closed-domain form used in this paper. We do not study the reflected model, near-zero recurrence, strict positivity, or boundary inaccessibility.

Assumption 2.1 (CEV exponents) Throughout this section we impose

$$\delta_1, \delta_2 \in [1/2, 1].$$

Let $x^+ = \max\{x, 0\}$. For $\delta \in [1/2, 1]$, the inequality

$$|(x^+)^\delta - (y^+)^\delta| \leq |x - y|^\delta, \quad x, y \in \mathbb{R},$$

implies that the positive-part diffusion coefficient $\sigma(x) = \xi(x^+)^\delta$ satisfies

$$|\sigma(x) - \sigma(y)| \leq \xi|x - y|^\delta, \quad x, y \in \mathbb{R}.$$

This is the Yamada-Watanabe modulus condition [16] in the one-dimensional form needed below. More explicitly, the condition requires a nondecreasing function ϱ with $\varrho(0) = 0$ and $\varrho(u) > 0$ for $u > 0$ such that

$$|\sigma(x) - \sigma(y)| \leq \varrho(|x - y|), \quad x, y \in \mathbb{R},$$

and

$$\int_0^\varepsilon \frac{du}{\varrho(u)^2} = \infty$$

for sufficiently small $\varepsilon > 0$. Here we may take $\varrho(u) = \xi u^\delta$, since

$$\int_0^\varepsilon \frac{du}{\varrho(u)^2} = \xi^{-2} \int_0^\varepsilon u^{-2\delta} du = \infty, \quad \delta \geq \frac{1}{2}.$$

The upper bound $\delta \leq 1$ gives at most linear growth.

The state space used for the GDMR dynamics is

$$\mathcal{D} = (0, \infty) \times [0, \infty)^2.$$

Theorem 2.2 (Strong well-posedness and nonnegativity) *Assume Assumption 2.1. For every $T > 0$ and every initial state*

$$S_0 > 0, \quad v_0 \geq 0, \quad v'_0 \geq 0,$$

the nonnegative-state system (1) admits a unique strong solution $(S_t, v_t, v'_t)_{t \in [0, T]}$ with continuous paths such that

$$S_t > 0, \quad v_t \geq 0, \quad v'_t \geq 0, \quad 0 \leq t \leq T, \quad \mathbb{Q}\text{-a.s.}$$

Moreover, for every deterministic initial state in $\mathcal{D}$, the solution is a time-homogeneous strong Markov process on $\mathcal{D}$.

Here, the boundary cases $v_0 = 0$ and $v'_0 = 0$ are included only to show that the variance system is well defined on the closed nonnegative domain and cannot leave it. They are not used as a modelling choice for calibration or numerical pricing. In the numerical part of the paper, the initial variance factors are strictly positive.

The proof is given in Appendix A. It follows the triangular structure of the model. First, the v' -equation is solved, then the v -equation is solved with v' as an adapted input, and finally, the asset equation is recovered from its stochastic exponential. The same one-way structure is used later in Subsection 3.2 and Proposition 3.1, where the Hybrid LSMC–PDE step conditions on the Brownian increments driving the variance factors.

The lower bound for CEV exponents, $\delta_i \geq 1/2$, is the Yamada-Watanabe threshold for pathwise uniqueness of power diffusion coefficients. We do not impose a Feller-type condition, since the theorem proves nonnegativity rather than strict positivity or boundary inaccessibility.

3 Bermudan option pricing

We now pass from the model dynamics to the one-step continuation calculation in the Bermudan recursion. The purpose of this section is to explain how the Hybrid LSMC–PDE method is used for the GDMR model and why the Brownian projection is needed. The Hybrid LSMC–PDE method was developed by Farahany, Jackson, and Jaimungal [7] for multidimensional stochastic volatility models. In this paper,

our contribution is to adapt that general framework to the GDMR model by deriving the Brownian projection and the resulting one-dimensional conditional log-price representation. We use the general hybrid-regression convergence theorem in [7]; the representation derived here provides the model-specific input needed to apply that theorem.

3.1 Bermudan stopping problem

Let

$$\pi = \{0 = t_0 < t_1 < \dots < t_M = T\}$$

be the Bermudan exercise grid. At t_n , the payoff is $h_n(S_{t_n})$, where $h_n : (0, \infty) \rightarrow \mathbb{R}$ is Borel. We assume

$$\mathbb{E} \left[\max_{0 \leq n \leq M} e^{-rt_n} |h_n(S_{t_n})| \right] < \infty.$$

We focus on Bermudan put options, with payoff $h_n(s) = (K - s)^+$, for illustrative purposes, since the early exercise feature is generally nontrivial. Under a nonnegative risk-free interest rate and a non-dividend-paying underlying asset, early exercise of call options is not optimal. Nevertheless, the methodology and discussion below can be readily adapted to Bermudan or American call options in settings where early exercise is possible, such as in the presence of dividends or negative interest rates.

By Theorem 2.2, the state process (S_t, v_t, v'_t) is a time-homogeneous Markov process on

$$\mathcal{D} = (0, \infty) \times [0, \infty)^2.$$

Thus the Bermudan value functions may be defined by backward induction. Write

$$\Delta t_n := t_{n+1} - t_n$$

and set the value function at maturity as

$$U_M(s, v, w) := h_M(s),$$

where w denotes the second variance coordinate. For $n = M - 1, \dots, 0$, define

$$C_n(s, v, w) := e^{-r\Delta t_n} \mathbb{E} \left[U_{n+1}(S_{t_{n+1}}, v_{t_{n+1}}, v'_{t_{n+1}}) \mid S_{t_n} = s, v_{t_n} = v, v'_{t_n} = w \right],$$

and following the dynamic programming principle, the value function at time t_n prior to maturity is

$$U_n(s, v, w) := \max\{h_n(s), C_n(s, v, w)\}.$$

The time-zero Bermudan value is

$$V_0 = U_0(S_0, v_0, v'_0).$$

3.2 Hybrid decomposition of the continuation value

A plain LSMC method approximates the continuation value in the Bermudan recursion by regressing simulated discounted next-step values on the current state. In a stochastic-volatility model this regression is more demanding than in a one-factor asset model, because the continuation value depends on both the asset level and the variance state. The Hybrid LSMC–PDE method reduces this regression problem by using the one-way coupling structure of the dynamics. The variance factors are simulated first; along each simulated variance path, the remaining conditional expectation in the asset direction is computed by a one-dimensional PDE step; the resulting pathwise conditional values are then regressed over the current variance state. The numerical realization of these steps, including the FFT evaluation and the regression matrices, is given in Section 4.

The idea can be written directly at the level of one Bermudan interval. Let F be a next-step value function; in the backward recursion, F is U_{n+1} or its numerical approximation. For a current state (s, v, w) , define

$$C_n^F(s, v, w) = e^{-r\Delta t_n} \mathbb{E} \left[F(S_{t_{n+1}}, v_{t_{n+1}}, v'_{t_{n+1}}) \mid S_{t_n} = s, v_{t_n} = v, v'_{t_n} = w \right].$$

The hybrid method evaluates this conditional expectation in two stages, using the one-way coupling of the model. The equations for (v, v') do not involve S , so over one exercise interval we may first simulate the Brownian increments driving the variance factors. Conditioning on these increments fixes the variance segment and the part of the asset Brownian motion correlated with the variance drivers. The only remaining asset randomness is the orthogonal Brownian component. This conditioning is used only inside the one-step conditional expectation and does not change the information used for the exercise decision at the exercise dates.

Let $\mathcal{H}_n$ denote this enlarged conditioning information; its precise definition is given in the next subsection. The inner, pathwise continuation value is

$$\bar{C}_n^F(s; \mathcal{H}_n) := e^{-r\Delta t_n} \mathbb{E} \left[F(S_{t_{n+1}}, v_{t_{n+1}}, v'_{t_{n+1}}) \mid S_{t_n} = s, \mathcal{H}_n \right]. \quad (2)$$

For each simulated variance path segment, $\bar{C}_n^F(\cdot; \mathcal{H}_n)$ is a function of the initial asset level. The original continuation value is recovered by the outer conditional expectation

$$C_n^F(s, v, w) = \mathbb{E} [\bar{C}_n^F(s; \mathcal{H}_n) \mid v_{t_n} = v, v'_{t_n} = w].$$

This expectation is the part approximated by least-squares regression over the current variance state.

To compute the continuation value, we must isolate the part of the asset Brownian motion that remains random after conditioning on $\mathcal{H}_n$. This is the role of the Brownian projection below. Once this residual component is identified, the conditional asset-price problem becomes one-dimensional in the log-price variable. The conditioning is only an iterated-expectation device and does not change the exercise information at time t_n .

3.3 Brownian projection

We now make the conditioning information in the preceding decomposition explicit. Fix an interval $[t_n, t_{n+1}]$ in the Bermudan exercise grid and define

$$\mathcal{H}_n := \mathcal{F}_{t_n} \vee \sigma\left(W_s^{(2)} - W_{t_n}^{(2)}, W_s^{(3)} - W_{t_n}^{(3)} : s \in [t_n, t_{n+1}]\right),$$

where $\mathcal{F}_1 \vee \mathcal{F}_2 := \sigma(\mathcal{F}_1 \cup \mathcal{F}_2)$.

Since v and v' are driven only by $W^{(2)}$ and $W^{(3)}$, the path $(v_s, v'_s)_{s \in [t_n, t_{n+1}]}$ is $\mathcal{H}_n$ -measurable. By assumption $|\rho_{23}| \neq 1$, we may project $W^{(1)}$ onto the span of $(W^{(2)}, W^{(3)})$. Set

$$\Sigma_{23} := \begin{bmatrix} 1 & \rho_{23} \\ \rho_{23} & 1 \end{bmatrix}, \quad \beta := \Sigma_{23}^{-1} \begin{bmatrix} \rho_{12} \\ \rho_{13} \end{bmatrix} = \begin{bmatrix} \beta_2 \\ \beta_3 \end{bmatrix}.$$

Equivalently,

$$\beta_2 = \frac{\rho_{12} - \rho_{13}\rho_{23}}{1 - \rho_{23}^2}, \quad \beta_3 = \frac{\rho_{13} - \rho_{12}\rho_{23}}{1 - \rho_{23}^2}. \quad (3)$$

The residual variance is

$$\begin{aligned} \sigma_\perp^2 &:= 1 - [\rho_{12} \ \rho_{13}] \Sigma_{23}^{-1} \begin{bmatrix} \rho_{12} \\ \rho_{13} \end{bmatrix} \\ &= \frac{1 - \rho_{12}^2 - \rho_{13}^2 - \rho_{23}^2 + 2\rho_{12}\rho_{13}\rho_{23}}{1 - \rho_{23}^2} \geq 0. \end{aligned} \quad (4)$$

When $\sigma_\perp > 0$,

$$B_t^\perp := \frac{W_t^{(1)} - \beta_2 W_t^{(2)} - \beta_3 W_t^{(3)}}{\sigma_\perp}$$

is a Brownian motion, and its increments over $[t_n, t_{n+1}]$ are independent of $\mathcal{H}_n$. Hence

$$dW_t^{(1)} = \beta_2 dW_t^{(2)} + \beta_3 dW_t^{(3)} + \sigma_\perp dB_t^\perp. \quad (5)$$

If $\sigma_\perp = 0$, the residual term is absent and the asset Brownian driver is fully determined by the Brownian drivers of the variance factors.

3.4 Conditional one-dimensional equation

Let $X_t := \log S_t$. Substituting Eq. (5) into the asset price equation for S_t in Eq. (1) gives

$$dX_t = \left(r - \frac{1}{2}v_t\right)dt + \sqrt{v_t}\left(\beta_2 dW_t^{(2)} + \beta_3 dW_t^{(3)}\right) + \sigma_\perp \sqrt{v_t} dB_t^\perp.$$

Define the $\mathcal{H}_n$ -measurable shift

$$Z_n := \int_{t_n}^{t_{n+1}} \sqrt{v_s} \left(\beta_2 dW_s^{(2)} + \beta_3 dW_s^{(3)}\right).$$

For $t \in [t_n, t_{n+1}]$, define also

$$I_n(t) := \int_t^{t_{n+1}} \left(r - \frac{1}{2}v_s\right) ds, \quad B_n(t) := \sigma_\perp^2 \int_t^{t_{n+1}} v_s ds,$$

and write $I_n := I_n(t_n)$ and $B_n := B_n(t_n)$.

Proposition 3.1 (Conditional one-step representation) *Let $G : \mathbb{R} \times [0, \infty)^2 \rightarrow \mathbb{R}$ be bounded and Borel. Conditionally on $\mathcal{H}_n$, the function*

$$u(t, y) := \mathbb{E} \left[G \left(Y_{t_{n+1}}^{t, y} + Z_n, v_{t_{n+1}}, v'_{t_{n+1}} \right) \middle| \mathcal{H}_n \right],$$

where

$$dY_s^{t, y} = \left(r - \frac{1}{2}v_s\right) ds + \sigma_\perp \sqrt{v_s} dB_s^\perp, \quad Y_t^{t, y} = y,$$

is the bounded mild solution of

$$\partial_t u + \left(r - \frac{1}{2}v_t\right) \partial_y u + \frac{1}{2} \sigma_\perp^2 v_t \partial_{yy} u = 0, \quad u(t_{n+1}, y) = G(y + Z_n, v_{t_{n+1}}, v'_{t_{n+1}}).$$

Equivalently, for an auxiliary $\xi \sim N(0, 1)$,

$$u(t, y) = \mathbb{E}_\xi \left[G \left(y + Z_n + I_n(t) + \sqrt{B_n(t)} \xi, v_{t_{n+1}}, v'_{t_{n+1}} \right) \right], \quad (6)$$

with the deterministic interpretation when $B_n(t) = 0$.

Proof After conditioning on $\mathcal{H}_n$, the variance path and the shift Z_n are fixed. If $\sigma_\perp > 0$, the only remaining randomness is the residual Brownian motion; if $\sigma_\perp = 0$, the conditional step is deterministic. The conditional Feynman-Kac formula gives the backward Kolmogorov equation. Since the coefficients are independent of y , the terminal log-price is Gaussian with mean $y + I_n(t)$ and variance $B_n(t)$, which gives the displayed formula. $\square$

For an asset-space next-step value function F and $G(y, v, w) := F(e^y, v, w)$, the pathwise quantity defined by Eq. (2) in Subsection 3.2 is

$$\bar{C}_n^F(s; \mathcal{H}_n) = e^{-r\Delta t_n} u(t_n, \log s).$$

Thus the conditional asset step depends on the sampled $(W^{(2)}, W^{(3)})$ -path over $[t_n, t_{n+1}]$ only through

$$Z_n, \quad I_n, \quad B_n, \quad v_{t_{n+1}}, \quad v'_{t_{n+1}},$$

while (v_{t_n}, v'_{t_n}) is the state used in the outer regression.

3.5 Connection with the hybrid convergence theorem

The almost-sure convergence theorem for the Hybrid LSMC–PDE regression is proved in Farahany, Jackson, and Jaimungal [7]. The preceding subsections identify the model-specific ingredients needed to apply that theorem.

First, the asset equation is multiplicative. For $0 \leq t < u \leq T$,

$$S_u = S_t R_{t,u},$$

where

$$R_{t,u} = \exp \left(\int_t^u \left(r - \frac{1}{2} v_s \right) ds + \int_t^u \sqrt{v_s} dW_s^{(1)} \right).$$

Thus the return factor $R_{t,u}$ depends on the Brownian and variance paths, but not on the value of S_t .

Second, Proposition 3.1 shows that, on each Bermudan interval, the sampled $(W^{(2)}, W^{(3)})$ -path enters the conditional asset-price step only through the finite-dimensional statistic

$$\Theta_n := (V_n, V_{n+1}, Z_n, I_n, B_n), \quad V_n = (v_{t_n}, v'_{t_n}).$$

Because the variance equations do not involve S , the model is one-way coupled. Hence we can apply the hybrid convergence theorem of [7, Theorem 1].

The remaining assumptions in the cited convergence theorem are stabilizing assumptions on the least-squares regression. In the numerical implementation below, these are enforced by working on the compact volatility rectangle $\mathcal{D}_v$ in Eq. (7), using a bounded volatility basis on that rectangle as in Eq. (8), and replacing the sample Gram inverse by the bounded inverse $[A_n^N]_R^{-1}$ in Eq. (10). The finite log-price grid $\mathcal{S}_h$ provides the corresponding compact asset domain for the conditional PDE step. Thus, the convergence theorem is applied to the same stabilized regression used in Section 4, rather than to a separate truncated algorithm.

Under these stabilized choices, and assuming the population Gram matrices are nonsingular, the hybrid regression coefficients converge almost surely by [7, Theorem 1]. Consequently, for each fixed asset level, the fitted continuation values converge uniformly on the chosen compact volatility domain.

4 Numerical implementation

The numerical experiments use the following implementation. The conditional representation is derived in Section 3; here we specify the grids, variance discretization, accumulated path statistics, FFT step, regression step, and time-zero estimator. We use this implementation because the conditional PDE admits a Gaussian structure in its solution, allowing the shift operation in the conditional formulation to be handled efficiently in Fourier space. In addition, the FFT implementation appears computationally advantageous in our numerical experiments.

4.1 Grids and basis functions

Let

$$\pi = \{0 = t_0 < t_1 < \dots < t_M = T\}$$

be the Bermudan exercise grid, and let h_n denote the exercise payoff at t_n . We work on a uniform log-price grid

$$y_i = y_{\min} + (i - 1)\Delta y, \quad \Delta y = \frac{y_{\max} - y_{\min}}{N_S - 1}, \quad i = 1, \dots, N_S,$$

and set

$$s_i = e^{y_i}, \quad \mathcal{S}_h := \{s_1, \dots, s_{N_S}\}.$$

The interval $[y_{\min}, y_{\max}]$ is chosen large enough to contain the relevant asset values used in the computation.

For the volatility variables, write $w = v'$. We work on the compact volatility rectangle

$$\mathcal{D}_v = [0, v_{\max}] \times [0, w_{\max}] \subset [0, \infty)^2. \quad (7)$$

The regression basis is chosen as a vector of continuous functions that are bounded on this rectangle,

$$\phi = (\phi_1, \dots, \phi_{d_B}), \quad \sup_{(v,w) \in \mathcal{D}_v} |\phi_k(v, w)| < \infty, \quad k = 1, \dots, d_B. \quad (8)$$

For the convergence statement, the numerical basis may be viewed as the restriction to $\mathcal{D}_v$ of compactly supported continuous functions on the full volatility state space; this extension does not affect the computed regression, since the regression is evaluated only on $\mathcal{D}_v$.

Once chosen, the domain $\mathcal{D}_v$ and the basis functions are fixed before the training regression.

At each exercise date, we approximate the continuation value by a separable regression form

$$c_n(s, v, w) \approx a_n(s) \cdot \phi(v, w), \quad a_n(s) \in \mathbb{R}^{d_B}.$$

Numerically, the coefficient vectors $a_n(s)$ are computed only at the grid points $s_i \in \mathcal{S}_h$. Off-grid asset values are evaluated by interpolation in the log-price coordinate.

4.2 Variance simulation and path statistics

Let

$$0 = \tau_0 < \tau_1 < \dots < \tau_L = T$$

be an Euler simulation grid containing all exercise dates in the Bermudan exercise grid π . Write

$$\Delta\tau_\ell := \tau_{\ell+1} - \tau_\ell.$$

For each path j , simulate correlated Gaussian increments

$$(\Delta W_\ell^{(2),j}, \Delta W_\ell^{(3),j})$$

with variances $\Delta\tau_\ell$ and covariance $\rho_{23}\Delta\tau_\ell$. To simulate the variance processes, we adopt a standard positivity-preserving truncated Euler scheme of the form:

$$\begin{aligned}\bar{w}_\ell^j &:= (w_\ell^j)^+, & \bar{v}_\ell^j &:= (v_\ell^j)^+, \\ w_{\ell+1}^j &:= \left[w_\ell^j + \kappa_2(\theta - \bar{w}_\ell^j)\Delta\tau_\ell + \xi_2(\bar{w}_\ell^j)^{\delta_2} \Delta W_\ell^{(3),j} \right]^+, \\ v_{\ell+1}^j &:= \left[v_\ell^j + \kappa_1(\bar{w}_\ell^j - \bar{v}_\ell^j)\Delta\tau_\ell + \xi_1(\bar{v}_\ell^j)^{\delta_1} \Delta W_\ell^{(2),j} \right]^+.\end{aligned}$$

For each exercise interval $[t_n, t_{n+1}]$, the path statistics are accumulated over the Euler substeps contained in that interval:

$$\begin{aligned}Z_n^j &:= \sum_{\tau_\ell \in [t_n, t_{n+1})} \sqrt{\bar{v}_\ell^j} \left(\beta_2 \Delta W_\ell^{(2),j} + \beta_3 \Delta W_\ell^{(3),j} \right), \\ I_n^j &:= \sum_{\tau_\ell \in [t_n, t_{n+1})} \left(r - \frac{1}{2} \bar{v}_\ell^j \right) \Delta\tau_\ell, \\ B_n^j &:= \sigma_\perp^2 \sum_{\tau_\ell \in [t_n, t_{n+1})} \bar{v}_\ell^j \Delta\tau_\ell.\end{aligned}$$

Here the projection coefficients β_2, β_3 and the residual volatility coefficient $\sigma_\perp$ are given by Eqs. (3) and (4), respectively. The left-point rule is natural for Z_n^j , since Z_n^j approximates a stochastic integral evaluated with left endpoints. The same left-point values are used for I_n^j and B_n^j for consistency with the Euler simulation, and the same increments $\Delta W_\ell^{(2),j}, \Delta W_\ell^{(3),j}$ are used in the variance updates and in Z_n^j .

4.3 Pathwise FFT propagation and regression

Fix $n \in \{0, \dots, M-1\}$. Here $\widehat{V}_{n+1}^N$ denotes the numerical Bermudan value surface already available from the next exercise date in the backward recursion. At maturity this surface is initialized by the payoff function

$$\widehat{V}_M^N(s_i, v, w) = h_M(s_i).$$

At earlier exercise dates it is the previously updated surface

$$\begin{aligned}\widehat{V}_{n+1}^N(s_i, v, w) &= \max\{h_{n+1}(s_i), \widehat{C}_{n+1}^N(s_i, v, w)\}, \\ \widehat{C}_{n+1}^N(s_i, v, w) &= a_{n+1}^N(s_i) \cdot \phi(v, w).\end{aligned}$$

Thus $\widehat{V}_{n+1}^N$ is already available before the following pathwise propagation step. For each Monte Carlo path j , the simulated Brownian increments driving the variance factors on $[t_n, t_{n+1}]$ determine the endpoint state

$$(v_{t_{n+1}}^j, w_{t_{n+1}}^j)$$

and the statistics Z_n^j, I_n^j, B_n^j . Define the terminal grid function

$$g_{n+1}^j(y_i) := \widehat{V}_{n+1}^N \left(e^{y_i}, v_{t_{n+1}}^j, w_{t_{n+1}}^j \right), \quad i = 1, \dots, N_S.$$

By the Gaussian representation Eq. (6) in Section 3,

$$u_n^j(y) = \mathbb{E}_\xi \left[g_{n+1}^j \left(y + Z_n^j + I_n^j + \sqrt{B_n^j} \xi \right) \right], \quad \xi \sim N(0, 1).$$

With the Fourier convention

$$\mathcal{F}f(\omega) = \int_{\mathbb{R}} e^{-i\omega y} f(y) dy,$$

we obtain

$$\mathcal{F}u_n^j(\omega) = \mathcal{F}g_{n+1}^j(\omega) \exp \left(i\omega Z_n^j + i\omega I_n^j - \frac{1}{2}\omega^2 B_n^j \right).$$

Define

$$\Psi_n^j(\omega) := i\omega Z_n^j + i\omega I_n^j - \frac{1}{2}\omega^2 B_n^j.$$

On the uniform log-price grid, the FFT implementation is

$$u_n^j(\cdot) \approx \text{FFT}^{-1} \left(\text{FFT}[g_{n+1}^j] \exp(\Psi_n^j) \right), \quad (9)$$

where the exponential is applied pointwise at the discrete Fourier frequencies. The FFT input is the unshifted terminal grid g_{n+1}^j ; the correlated shift Z_n^j is included only through the multiplier. Equivalently, one may shift the terminal grid first and omit $i\omega Z_n^j$, but the two conventions must not be combined. If $B_n^j = 0$, the conditional propagation reduces to the deterministic log-price shift $Z_n^j + I_n^j$.

The pathwise pre-surface on the asset grid is

$$\widehat{C}_n^j(s_i) := e^{-r\Delta t_n} u_n^j(y_i), \quad i = 1, \dots, N_S.$$

The regression step is performed at each exercise date t_n , $n = M-1, \dots, 1$, and at each asset grid point s_i . It estimates the dependence of the pathwise pre-surface $\widehat{C}_n^j(s_i)$ on the current volatility state. Define

$$V_n^j := (v_{t_n}^j, w_{t_n}^j).$$

The sample Gram matrix is

$$A_n^N := \frac{1}{N} \sum_{j=1}^N \phi(V_n^j) \phi(V_n^j)^\top,$$

and the right-hand side for the grid point s_i is

$$b_n^N(s_i) := \frac{1}{N} \sum_{j=1}^N \phi(V_n^j) \widehat{C}_n^j(s_i).$$

For numerical stability, we use the truncated inverse

$$[A]_R^{-1} := \begin{cases} A^{-1}, & \text{if } A \text{ is invertible and } \|A^{-1}\| \leq R, \\ 0, & \text{otherwise.} \end{cases} \quad (10)$$

Then set

$$a_n^N(s_i) := [A_n^N]_R^{-1} b_n^N(s_i). \quad (11)$$

The fitted continuation surface is

$$\widehat{C}_n^N(s_i, v, w) := a_n^N(s_i) \cdot \phi(v, w),$$

and the updated price for the Bermudan option is

$$\widehat{V}_n^N(s_i, v, w) := \max \left\{ h_n(s_i), \widehat{C}_n^N(s_i, v, w) \right\}.$$

Starting from

$$\widehat{V}_M^N(s_i, v, w) := h_M(s_i),$$

the recursion is performed backward for $n = M - 1, \dots, 1$. The fitted functions are defined at the asset grid points and extended to off-grid asset values by log-price interpolation.

4.4 Time-zero estimator and pseudocode

At $t_0 = 0$, the initial volatility state is fixed, so no regression is needed. After $\widehat{V}_1^N$ has been fitted, the time-zero continuation value is evaluated on the asset grid by averaging first-step pathwise pre-surfaces:

$$\widehat{C}_0^N(s_i) := \frac{1}{N} \sum_{j=1}^N \widehat{C}_0^j(s_i).$$

The averaged grid values define $\widehat{C}_0^N$ on the asset grid; off-grid values are evaluated by the log-price interpolation convention stated above. Thus $\widehat{C}_0^N(S_0)$ denotes the averaged continuation value at the initial asset price, and the Hybrid LSMC–PDE estimator is

$$\widehat{V}_0^N(S_0) = \max \left\{ h_0(S_0), \widehat{C}_0^N(S_0) \right\}. \quad (12)$$

Note that the present implementation has the advantage that option prices are computed simultaneously for multiple initial stock prices on the stock-price grid,

equivalently for multiple moneyness levels since the strike price K is fixed, which is particularly desirable for calibration. In contrast, a plain LSMC approach typically produces the option price corresponding to a single initial stock price in each run.

The implementation of the Hybrid LSMC–PDE method is summarized in Algorithm 1.

Algorithm 1 Hybrid LSMC–PDE recursion

Require: Exercise grid $\{t_n\}_{n=0}^M$, Euler grid containing the exercise dates, log-price grid $\{y_i\}_{i=1}^{N_S}$, volatility domain $\mathcal{D}_v$, basis ϕ , truncation level R , and path count N

Ensure: Hybrid LSMC–PDE estimator $\hat{V}_0^N(S_0)$ and fitted continuation surfaces $\hat{C}_n^N$

- 1: Simulate N training paths of the variance factors on the Euler grid.
 - 2: Store exercise-date volatility states and accumulate Z_n^j , I_n^j , and B_n^j over each interval $[t_n, t_{n+1}]$.
 - 3: Set $\hat{V}_M^N(s_i, v, w) \leftarrow h_M(s_i)$ on $\mathcal{S}_h \times \mathcal{D}_v$.
 - 4: **for** $n = M - 1, \dots, 1$ **do**
 - 5: **for** $j = 1, \dots, N$ **do**
 - 6: Compute $\{\hat{C}_n^j(s_i)\}_{i=1}^{N_S}$ using the unshifted terminal grid g_{n+1}^j in Eq. (9), with Z_n^j included only through the multiplier $\exp(\Psi_n^j)$; do not pre-shift g_{n+1}^j by Z_n^j .
 - 7: **end for**
 - 8: **for** each $s_i \in \mathcal{S}_h$ **do**
 - 9: Compute $b_n^N(s_i)$ and $a_n^N(s_i)$ by the truncated-inverse regression Eq. (11).
 - 10: **end for**
 - 11: Set $\hat{C}_n^N(s_i, v, w) \leftarrow a_n^N(s_i) \cdot \phi(v, w)$ for all $s_i \in \mathcal{S}_h$.
 - 12: Set $\hat{V}_n^N(s_i, v, w) \leftarrow \max\{h_n(s_i), \hat{C}_n^N(s_i, v, w)\}$ for all $s_i \in \mathcal{S}_h$.
 - 13: **end for**
 - 14: Compute the first-step pre-surfaces $\hat{C}_0^j(s_i)$ for $j = 1, \dots, N$ using the simulated paths on $[t_0, t_1]$.
 - 15: Set $\hat{C}_0^N(s_i) \leftarrow N^{-1} \sum_{j=1}^N \hat{C}_0^j(s_i)$.
 - 16: Evaluate $\hat{C}_0^N(S_0)$ from the averaged continuation grid at $y = \log S_0$, using linear interpolation if needed.
 - 17: **return** $\hat{V}_0^N(S_0) = \max\{h_0(S_0), \hat{C}_0^N(S_0)\}$.
-

5 Numerical experiments

For numerical illustration, we consider Bermudan put pricing experiments using the fixed parameter set in Table 1. Since no closed-form solution is available for Bermudan option prices under the GDMR model, we compare the plain LSMC method [2] and the Hybrid LSMC–PDE estimator in Eq. (12) against large-simulation LSMC reference prices using relative errors computed with respect to these reference prices.

Although the reference prices are not exact solutions, they serve as numerical benchmarks throughout the paper. For simplicity, we refer to these benchmark-relative deviations as relative errors.

5.1 Experimental setting and reference values

Table 1 reports the fixed GDMR parameter set and numerical settings used in the experiments.

Table 1 Parameter set and numerical settings used in the pricing experiments.

Symbol	Value	Interpretation
S_0	100	initial asset price
v_0	0.114	initial instantaneous variance factor
v'_0	0.110	initial secondary variance factor
r	0.02	risk-free rate
T	1.0	maturity in years
κ_1	5.5	mean-reversion speed of v_t toward v'_t
κ_2	0.1	mean-reversion speed of v'_t toward θ
θ	0.078	long-run variance level
ξ_1	2.689	volatility scale of the first variance factor
ξ_2	0.502	volatility scale of the second variance factor
δ_1	0.94	CEV exponent of the first variance factor
δ_2	0.94	CEV exponent of the second variance factor
ρ_{12}	-0.982	correlation between the first and second Brownian drivers
ρ_{13}	-0.727	correlation between the first and third Brownian drivers
ρ_{23}	0.590	correlation between the second and third Brownian drivers
<i>Fixed numerical setting</i>		
N_{ex}	12	exercise dates after t_0 , including maturity
K	{70, 80, 90, 100, 110}	strike prices used in all experiments
N_S	301	asset-grid points in the conditional PDE solver
$(y_{\min}, y_{\max})$	$(\log(0.30S_0), \log(3.5 \max(S_0, K)))$	lower and upper log-price grid bounds

A volatility truncation quantile $q_v = 0.999$ is used to choose the bounds $v_{\max}$ and $w_{\max}$ in the compact volatility domain $\mathcal{D}_v$ in Eq. (7). Thus, $v_{\max}$ and $w_{\max}$ are chosen so that approximately 99.9% of the simulated values of v and v' at the exercise dates lie below the corresponding bounds, truncating only the upper 0.1% tail. The Hybrid LSMC–PDE grid and truncation settings are kept fixed throughout all experiments.

The mean-reversion, volatility-scale, initial-variance, and correlation parameters are taken from the calibrated double mean-reverting specification of Bayer, Gatheral and Karlsmark [12]. Their calibration assumes $r = 0$. In the present Bermudan pricing experiments we keep those variance and correlation parameters, set $r = 0.02$, and use $\delta_1 = \delta_2 = 0.94$ for the CEV-type generalization. Thus the numerical section is not a new calibration study; it uses a parameter set from the double mean-reverting volatility literature as a fixed test case. The correlation assumptions used in Section 3 are satisfied. In particular, $|\rho_{23}| < 1$ and $\rho_{1,2}, \rho_{1,3} \in [-1, 0]$.

The plain LSMC regressions use a cubic 16-term basis

$$1, x, y, z, x^2, y^2, z^2, xy, xz, yz, x^3, y^3, z^3, p, p^2, p^3,$$

where $x = S/S_0$, $y = v/\theta$, $z = v'/\theta$, and $p = (K - S)^+/K$. The hybrid regressions use a volatility-only cubic basis

$$1, \tilde{y}, \tilde{z}, \tilde{y}^2, \tilde{y}\tilde{z}, \tilde{z}^2, \tilde{y}^3, \tilde{y}^2\tilde{z}, \tilde{y}\tilde{z}^2, \tilde{z}^3,$$

where $\tilde{y} = \min(v, v_{\max})/v_{\max}$ and $\tilde{z} = \min(v', v'_{\max})/v'_{\max}$.

The reference prices, denoted by V^{ref} and reported in Table 2, are large-simulation plain LSMC estimates computed using 1200 Euler steps and 1,200,000 simulation paths. The table also reports empirical Monte Carlo standard errors and associated 95% confidence intervals. In subsequent relative-error calculations and plots, only V^{ref} is used as the benchmark, while the associated standard errors and confidence intervals are included for information only.

For an estimator V and reference price V^{ref} , we report the relative error

$$\varepsilon_{\text{rel}} := \left| \frac{V - V^{\text{ref}}}{V^{\text{ref}}} \right|,$$

where V^{ref} is treated as fixed. To visualize the statistical variability of the estimators in the relative-error space, error bars in the relative-error plots are obtained by mapping the 95% confidence intervals $[V_{\text{left}}, V_{\text{right}}]$ of the estimators through

$$V \mapsto \left| \frac{V - V^{\text{ref}}}{V^{\text{ref}}} \right|.$$

Table 2 LSMC reference values for the parameter set in Table 1.

K	Reference price (V^{ref})	Std. error	95% confidence interval
70	2.522851	0.007103	[2.508929, 2.536773]
80	4.298933	0.009345	[4.280617, 4.317249]
90	6.942119	0.011774	[6.919042, 6.965196]
100	10.682037	0.014105	[10.654391, 10.709683]
110	15.743520	0.015757	[15.712636, 15.774404]

5.2 Fixed path budget and varying Euler steps

The first experiment keeps the Monte Carlo path budget fixed and varies the number of Euler time steps. The two path budgets are 20,000 and 60,000, and both methods are evaluated at 24, 48, 72, and 96 Euler steps. This experiment tests the sensitivity of the estimators to the Euler discretization.

For 20,000 paths, the relative errors of the plain LSMC estimator over the full step-sweep grid range from 0.344% to 8.645%, while those of the Hybrid LSMC-PDE estimator range from 0.019% to 7.459%. Figure 1 reports the strike-wise relative errors.

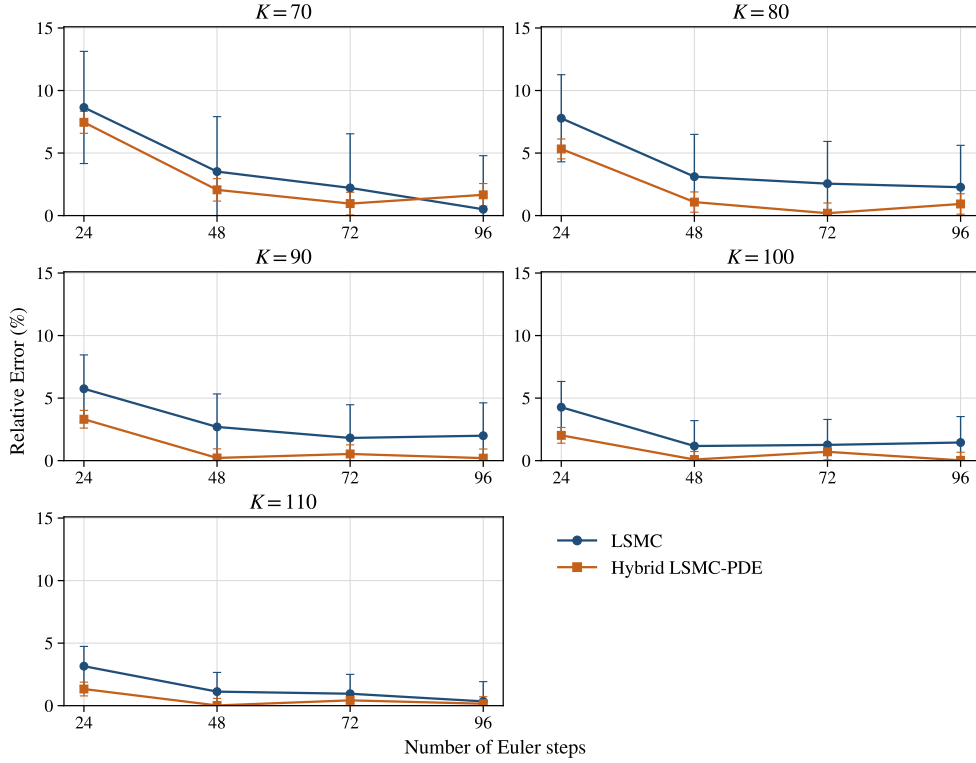


Fig. 1 Relative errors with associated error bars for 20,000-path step-sweep experiment.

For 60,000 paths, the same design gives relative errors of the plain LSMC estimator ranging from 0.102% to 7.720%, while those of the Hybrid LSMC–PDE estimator range from 0.028% to 7.857%. Figure 2 reports the corresponding strike-wise relative errors. The Hybrid LSMC–PDE estimator does not give a smaller relative error in every step-count configuration, but it gives smaller relative errors in most of the reported configurations, with the clearest differences at the intermediate step levels.

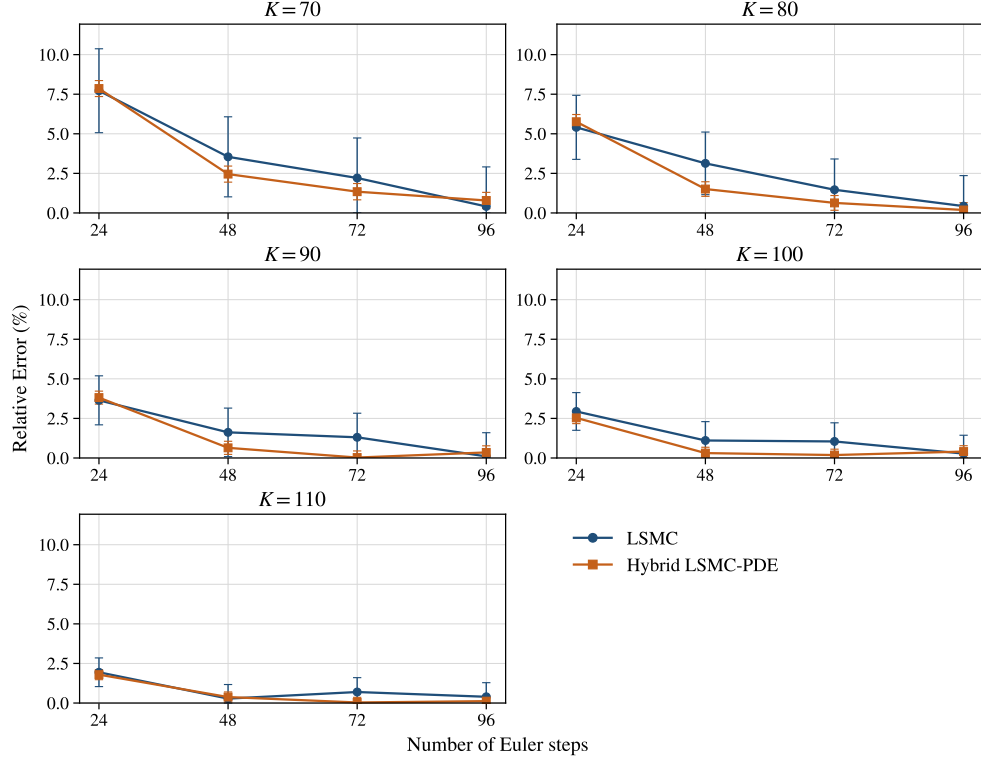


Fig. 2 Relative errors with associated error bars for 60,000-path step-sweep experiment.

5.3 Fixed Euler step count and varying path budget

The second experiment fixes the number of Euler time steps and varies the Monte Carlo path budget. The reported path counts are 250, 1000, 5000, 10000, 20000, 40000, and 60000. Two discretization levels, 48 and 60 Euler steps, are considered. This experiment focuses on sampling variation and regression error.

At 48 Euler steps, the at-the-money strike $K = 100$ gives relative errors of the plain LSMC estimator ranging from 1.105% to 25.948% over the path grid, while those of the Hybrid LSMC–PDE estimator range from 0.088% to 11.118%. Figure 3 shows the corresponding path-count sweep for all five strikes.

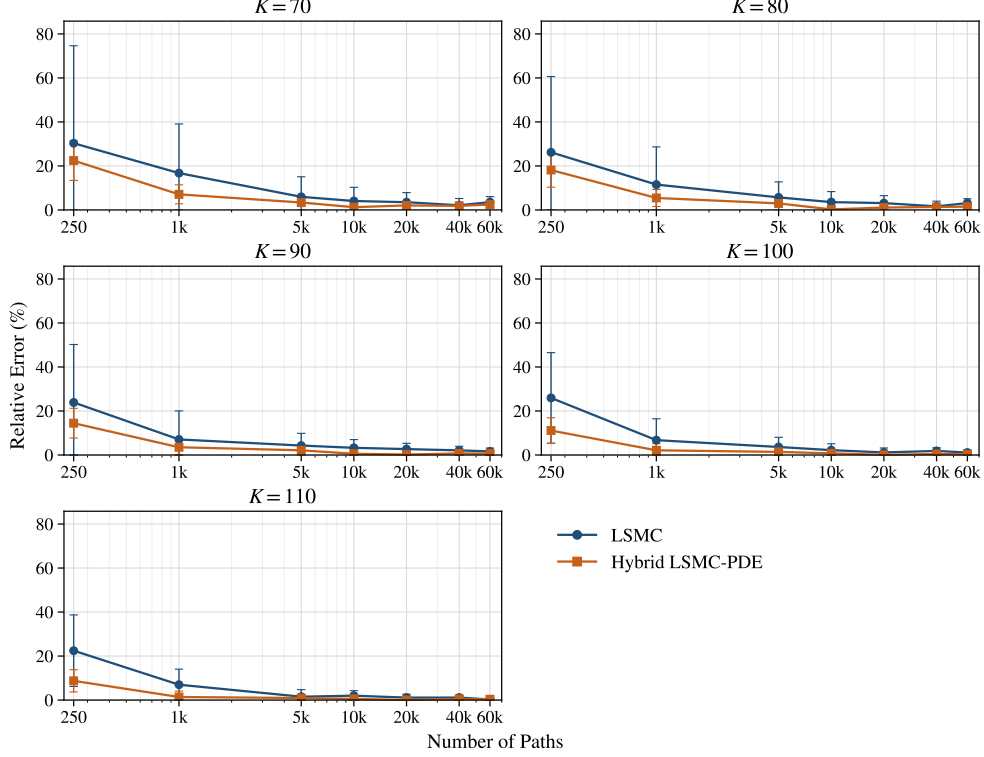


Fig. 3 Relative errors with associated error bars for the fixed 48-step path-sweep experiment.

At 60 Euler steps, the corresponding relative errors of the plain LSMC estimator range from 1.324% to 28.932%, while those of the Hybrid LSMC–PDE estimator range from 0.049% to 10.102%. Figure 4 reports the corresponding path-count sweep for all five strikes. The reduction in relative error is most visible in the low- and moderate-path regimes, where the plain LSMC estimator is most affected by sampling noise.

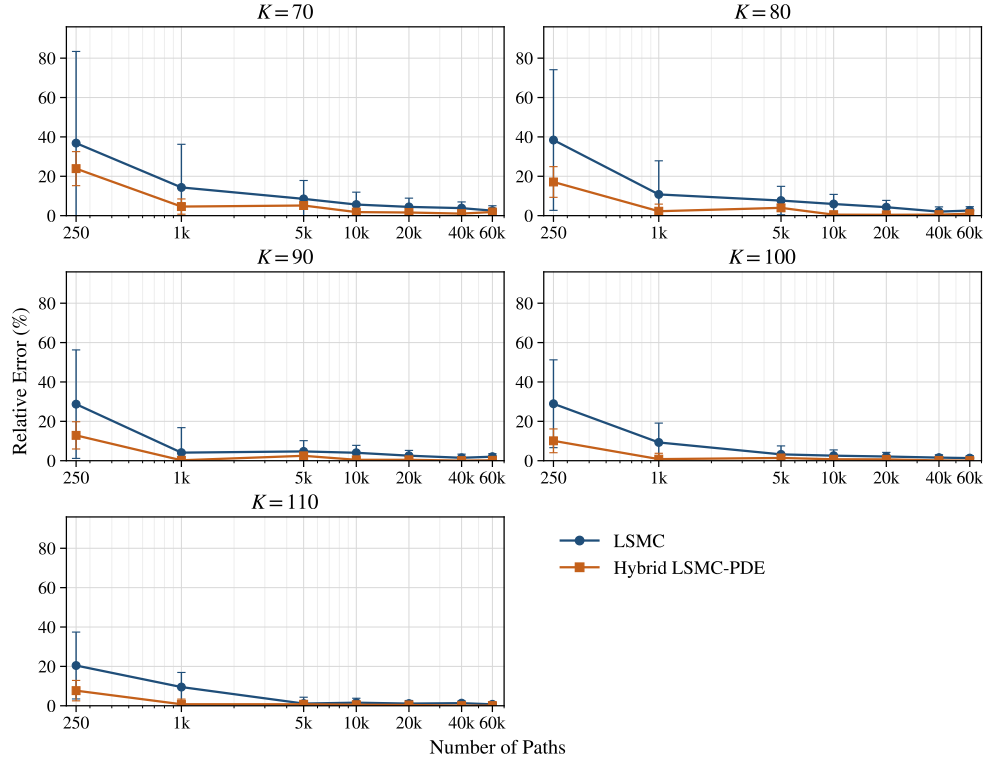


Fig. 4 Relative errors with associated error bars for the fixed 60-step path-sweep experiment.

As shown in Figures 1–4, the error bars are generally largest in the low-path experiments, especially for the plain LSMC estimator. In comparison, the Hybrid LSMC–PDE estimator is usually centered at lower relative errors and shows more stable error bars in the low and moderate path regimes. As the number of paths increases, the error bars shrink for both methods, and the difference between plain LSMC and Hybrid LSMC–PDE becomes smaller.

Overall, the experimental results are consistent with the conditional PDE step reducing the dimension of the continuation regression. Note that the comparisons are made against LSMC reference prices in Table 2 obtained from large-scale simulations rather than exact prices. Since these benchmarks are naturally aligned with the LSMC methodology, the competitive performance of the Hybrid LSMC–PDE method is encouraging. One possible explanation is that, by exploiting model structure through a conditional PDE for the stock-price dependence rather than a full regression approximation of the continuation value, the Hybrid LSMC–PDE method may provide a more faithful representation of the exercise decision.

The complete set of estimated option prices is reported in Appendix B and may serve as benchmark values for future studies. More specifically, the pricing estimates underlying Figures 1–4 are reported in Tables B1–B4, respectively.

6 Conclusion

In this paper, we studied the pricing of Bermudan options under a GDMR model. Bermudan pricing requires the approximation of continuation values at each exercise date, and the two variance factors in the GDMR model make this regression problem more difficult than in a one-factor model. At the same time, the model has a useful one-way coupling; the variance factors can be simulated independently of the stock price. This structure leads naturally to the Hybrid LSMC–PDE method, since the stock-price direction can be handled conditionally while the remaining variance dependence is approximated by least-squares regression. We condition on the Brownian paths driving the variance factors and project the asset Brownian motion onto these drivers. Under this conditioning, each continuation step reduces to a one-dimensional problem in log-price. This gives the Gaussian conditional representation based on Z_n, I_n, B_n , and the variance values at the next exercise date, and leads to an FFT-based implementation of the conditional asset step. We also included that, for the CEV exponents $\delta_1, \delta_2 \in [1/2, 1]$, the GDMR variance system has a unique strong solution and remains nonnegative. This property is needed since the construction of Hybrid LSMC–PDE relies on valid simulated variance paths.

We conducted the numerical study using a set of calibrated parameters from the literature. We compared plain LSMC with the Hybrid LSMC–PDE estimator across different Euler step counts and different numbers of Monte Carlo paths. Because no closed-form Bermudan benchmark is available for the GDMR model, the comparison was made against large-simulation plain LSMC reference prices. Within this setup, the Hybrid LSMC–PDE estimator produced smaller relative errors in many of the reported cases. The clearest gains appeared when the number of paths was low or moderate. The error bars also support this observation: they are generally largest in

the low-path experiments, especially for the plain LSMC estimator, while the Hybrid LSMC–PDE estimator shows more stable error bars in the low and moderate path regimes.

These results suggest that the conditional PDE step may reduce the difficulty in approximating the continuation value in this model. For future work, we can test the method under additional calibrated parameter sets, consider alternative reference prices or bias-control procedures such as low-biased or dual estimators, and study other payoff types and early-exercise contracts.

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Declarations

Funding

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Conflict of interest/Competing interests

The authors declare that they have no competing interests.

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Data availability

No external datasets were used in this study. All numerical results were generated from simulated data based on the model specification described in the paper.

Materials availability

Not applicable.

Code availability

The computational code used for the numerical experiments in this study is available from the corresponding author upon reasonable request.

Author contributions

MDK and YN contributed to the conceptualization, mathematical analysis, numerical design, interpretation of results, and manuscript preparation. All authors read and approved the final manuscript.

Appendix A Proof of Theorem 2.2

Proof. We first work with auxiliary equations on $\mathbb{R}$. For this purpose, the variance diffusion coefficients are extended by

$$\sigma_i(x) = \xi_i(x^+)^{\delta_i}, \quad x^+ = \max\{x, 0\}, \quad i = 1, 2.$$

Once nonnegativity is proved, this extension is inactive because both variance factors stay in $[0, \infty)$. The auxiliary equations then coincide with the original system on the state space $\mathcal{D}$.

Step 1: construction of v' . Define

$$b_2(x) := \kappa_2(\theta - x), \quad \sigma_2(x) := \xi_2(x^+)^{\delta_2}, \quad x \in \mathbb{R}.$$

The drift b_2 is globally Lipschitz and has linear growth. The diffusion coefficient σ_2 is continuous, has at most linear growth, and satisfies the Yamada-Watanabe modulus condition by Assumption 2.1. Therefore pathwise uniqueness holds for

$$dv'_t = b_2(v'_t)dt + \sigma_2(v'_t)dW_t^{(3)}, \quad v'_0 \geq 0;$$

see [16] and [19, Ch. 5]. Since the coefficients are continuous with linear growth, weak existence holds. Weak existence together with pathwise uniqueness yields a unique strong solution by the Yamada-Watanabe theorem.

It remains to show that the solution does not leave the nonnegative half-line. At the boundary,

$$\sigma_2(0) = 0, \quad b_2(0) = \kappa_2\theta \geq 0.$$

Moreover, on the negative extension one has

$$\sigma_2(x) = 0, \quad b_2(x) = \kappa_2(\theta - x) \geq 0, \quad x \leq 0.$$

Thus the diffusion has no outward noise at the boundary and the drift points inward on the negative side. The standard one-dimensional stochastic invariance criterion for closed intervals therefore yields

$$v'_t \geq 0, \quad 0 \leq t \leq T, \quad \mathbb{Q}\text{-a.s.}$$

Step 2: construction of v . Having constructed v' , define

$$b_1(t, x) := \kappa_1(v'_t - x), \quad \sigma_1(x) := \xi_1(x^+)^{\delta_1}.$$

For $m \geq 1$, set

$$\tau_m := \inf\{t \geq 0 : v'_t > m\} \wedge T.$$

Since v' has continuous paths on $[0, T]$, we have $\tau_m \uparrow T$ almost surely. On $[0, \tau_m]$, the drift b_1 is progressively measurable, Lipschitz in x with constant κ_1 , and satisfies

$$|b_1(t, x)| \leq \kappa_1(m + |x|).$$

The diffusion coefficient σ_1 is continuous, has at most linear growth, and satisfies the Yamada-Watanabe modulus condition. The localized one-dimensional existence and pathwise-uniqueness theorem for SDEs with progressive random coefficients, obtained by the same Yamada-Watanabe argument after localization, therefore gives a unique strong stopped solution of

$$dv_t = b_1(t, v_t)dt + \sigma_1(v_t)dW_t^{(2)}, \quad t \leq \tau_m.$$

The possible correlation between $W^{(2)}$ and $W^{(3)}$ is immaterial here: pathwise uniqueness compares solutions driven by the same Brownian vector and the same realized input path v' . By pathwise uniqueness, the stopped solutions are compatible on overlaps. Letting $m \rightarrow \infty$ gives a unique strong solution for v on $[0, T]$.

The same invariance argument gives nonnegativity. Since $v'_t \geq 0$,

$$b_1(t, 0) = \kappa_1 v'_t \geq 0, \quad \sigma_1(0) = 0.$$

On the negative extension,

$$b_1(t, x) = \kappa_1(v'_t - x) \geq 0, \quad \sigma_1(x) = 0, \quad x \leq 0.$$

Hence the closed half-line is invariant and

$$v_t \geq 0, \quad 0 \leq t \leq T, \quad \mathbb{Q}\text{-a.s.}$$

Step 3: construction of S . The process v is continuous and finite on the compact interval $[0, T]$. Therefore

$$\int_0^T v_s ds < \infty \quad \mathbb{Q}\text{-a.s.}$$

The asset equation is linear in S , and its unique strong solution is

$$S_t = S_0 \exp \left(\int_0^t \left(r - \frac{1}{2} v_s \right) ds + \int_0^t \sqrt{v_s} dW_s^{(1)} \right).$$

Thus $S_t > 0$ for all $t \in [0, T]$ whenever $S_0 > 0$.

The auxiliary positive-part extension is inactive along the constructed variance paths, because $v_t, v'_t \geq 0$. Hence the constructed solution solves the original nonnegative-state system (1). Pathwise uniqueness for the auxiliary triangular system gives pathwise uniqueness for (1).

Finally, existence and pathwise uniqueness for every deterministic initial state imply uniqueness in law. Since the coefficients are time homogeneous, the standard strong Markov property for well-posed time-homogeneous SDEs gives that the solution is a time-homogeneous strong Markov process on $\mathcal{D}$. $\square$

Appendix B Pricing estimates for the numerical experiments

This appendix reports the Bermudan put pricing estimates corresponding to the numerical experiments in Section 5. The pricing estimates reported in Tables B1–B4 correspond to the numerical experiments in Figures 1–4. The associated large-simulation LSMC reference prices are reported in Table 2. All entries in Tables B1–B4 are rounded to three decimal places. The label LSMC denotes the plain least-squares Monte Carlo estimator, and Hybrid denotes the Hybrid LSMC–PDE estimator.

Table B1 Pricing estimates in the fixed-path step-sweep experiment with 20,000 paths. Prices are rounded to three decimal places.

Method/step	24	48	72	96
LSMC $K = 70$	2.741	2.612	2.579	2.536
Hybrid $K = 70$	2.711	2.575	2.547	2.565
LSMC $K = 80$	4.634	4.433	4.409	4.396
Hybrid $K = 80$	4.528	4.345	4.307	4.339
LSMC $K = 90$	7.341	7.129	7.068	7.080
Hybrid $K = 90$	7.172	6.957	6.905	6.956
LSMC $K = 100$	11.138	10.807	10.816	10.837
Hybrid $K = 100$	10.898	10.673	10.607	10.679
LSMC $K = 110$	16.241	15.920	15.894	15.798
Hybrid $K = 110$	15.953	15.747	15.676	15.768

Table B2 Pricing estimates in the fixed-path step-sweep experiment with 60,000 paths. Prices are rounded to three decimal places.

Method/step	24	48	72	96
LSMC $K = 70$	2.718	2.612	2.579	2.533
Hybrid $K = 70$	2.721	2.585	2.557	2.543
LSMC $K = 80$	4.531	4.434	4.362	4.318
Hybrid $K = 80$	4.547	4.364	4.326	4.307
LSMC $K = 90$	7.195	7.055	7.033	6.949
Hybrid $K = 90$	7.207	6.987	6.940	6.918
LSMC $K = 100$	10.996	10.800	10.794	10.712
Hybrid $K = 100$	10.953	10.715	10.662	10.639
LSMC $K = 110$	16.050	15.788	15.853	15.806
Hybrid $K = 110$	16.026	15.803	15.749	15.725

Table B3 Pricing estimates in the fixed 48-step path-sweep experiment. Prices are rounded to three decimal places.

Method/path count	250	1,000	5,000	10,000	20,000	40,000	60,000
LSMC $K = 70$	3.288	2.946	2.673	2.625	2.612	2.576	2.612
Hybrid $K = 70$	3.089	2.702	2.608	2.554	2.575	2.570	2.585
LSMC $K = 80$	5.425	4.795	4.545	4.452	4.433	4.369	4.434
Hybrid $K = 80$	5.079	4.533	4.427	4.310	4.345	4.357	4.364
LSMC $K = 90$	8.601	7.433	7.241	7.169	7.129	7.088	7.055
Hybrid $K = 90$	7.946	7.187	7.090	6.903	6.957	6.992	6.987
LSMC $K = 100$	13.454	11.404	11.072	10.918	10.807	10.881	10.800
Hybrid $K = 100$	11.870	10.910	10.833	10.601	10.673	10.730	10.715
LSMC $K = 110$	19.278	16.843	15.985	16.057	15.920	15.921	15.788
Hybrid $K = 110$	17.120	15.962	15.889	15.660	15.747	15.819	15.803

Table B4 Pricing estimates in the fixed 60-step path-sweep experiment. Prices are rounded to three decimal places.

Method/path count	250	1,000	5,000	10,000	20,000	40,000	60,000
LSMC $K = 70$	3.452	2.884	2.738	2.665	2.635	2.619	2.587
Hybrid $K = 70$	3.125	2.639	2.653	2.568	2.563	2.549	2.570
LSMC $K = 80$	5.950	4.762	4.629	4.553	4.484	4.387	4.411
Hybrid $K = 80$	5.034	4.396	4.470	4.321	4.318	4.321	4.342
LSMC $K = 90$	8.936	7.227	7.270	7.222	7.120	7.045	7.084
Hybrid $K = 90$	7.836	6.957	7.115	6.910	6.910	6.939	6.957
LSMC $K = 100$	13.773	11.673	11.026	10.955	10.910	10.850	10.823
Hybrid $K = 100$	11.761	10.596	10.831	10.604	10.610	10.666	10.677
LSMC $K = 110$	18.962	17.240	15.920	15.991	15.918	15.952	15.860
Hybrid $K = 110$	16.953	15.621	15.863	15.659	15.680	15.755	15.757

References

- [1] Tsitsiklis, J.N., Van Roy, B.: Regression methods for pricing complex American-style options. *IEEE Transactions on Neural Networks* **12**(4), 694–703 (2001) <https://doi.org/10.1109/72.935083>
- [2] Longstaff, F.A., Schwartz, E.S.: Valuing American Options by Simulation: A Simple Least-Squares Approach. *The Review of Financial Studies* **14**(1), 113–147 (2001) <https://doi.org/10.1093/rfs/14.1.113>
- [3] Clément, E., Lamberton, D., Protter, P.: An analysis of a least squares regression method for American option pricing. *Finance and Stochastics* **6**(4), 449–471 (2002) <https://doi.org/10.1007/s007800200071>
- [4] Loeper, G., Pironneau, O.: A mixed PDE/Monte-Carlo method for stochastic volatility models. *Comptes Rendus Mathématique* **347**(9-10), 559–563 (2009) <https://doi.org/10.1016/j.crma.2009.02.021>
- [5] Lipp, T., Loeper, G., Pironneau, O.: Mixing Monte-Carlo and partial differential equations for pricing options. *Chinese Annals of Mathematics, Series B* **34**(2), 255–276 (2013) <https://doi.org/10.1007/s11401-013-0763-2>
- [6] Cozma, A., Reisinger, C.: A Mixed Monte Carlo and partial differential equation variance reduction method for foreign exchange options under the Heston-Cox-Ingersoll-Ross model. *The Journal of Computational Finance* **20**(3), 109–149 (2017) <https://doi.org/10.21314/JCF.2016.318>
- [7] Farahany, D., Jackson, K.R., Jaimungal, S.: Mixing LSMC and PDE Methods to Price Bermudan Options. *SIAM Journal on Financial Mathematics* **11**(1), 201–239 (2020) <https://doi.org/10.1137/19M1249035>
- [8] Gatheral, J.: Consistent Modelling of SPX and VIX Options. Bachelier Congress, London. Conference presentation (2008)
- [9] Heston, S.L.: A Closed-Form Solution for Options with Stochastic Volatility with

- Applications to Bond and Currency Options. *The Review of Financial Studies* **6**(2), 327–343 (1993) <https://doi.org/10.1093/rfs/6.2.327>
- [10] Cox, J.C., Ingersoll, J.E., Ross, S.A.: A Theory of the Term Structure of Interest Rates. *Econometrica* **53**(2), 385–407 (1985) <https://doi.org/10.2307/1911242>
 - [11] Gatheral, J.: *The Volatility Surface: A Practitioner’s Guide*. John Wiley & Sons, Hoboken, NJ (2006). <https://www.wiley.com/en-us/The%2BVolatility%2BSurface%3A%2BA%2BPractitioner%27s%2BGuide-p-9780471792512>
 - [12] Bayer, C., Gatheral, J., Karlsmark, M.: Fast Ninomiya-Victoir calibration of the double-mean-reverting model. *Quantitative Finance* **13**(11), 1813–1829 (2013) <https://doi.org/10.1080/14697688.2013.818245>
 - [13] Haida, B., Dimitrov, M.K., Marmaras, T., Ni, Y.: Almost exact mixed scheme for gatheral double mean reverting model. In: Silvestrov, S., Malyarenko, A. (eds.) *Mathematical Structures I - Stochastic and Analytic Structures with Applications*. Springer Proceedings in Mathematics and Statistics, vol. 532. Springer, Cham (2026). Accepted for publication. <https://link.springer.com/book/9783032213648>
 - [14] Cox, J.C.: Notes on Option Pricing I: Constant Elasticity of Variance Diffusions. Working paper, Stanford University (1975)
 - [15] Chan, K.C., Karolyi, G.A., Longstaff, F.A., Sanders, A.B.: An Empirical Comparison of Alternative Models of the Short-Term Interest Rate. *The Journal of Finance* **47**(3), 1209–1227 (1992) <https://doi.org/10.1111/j.1540-6261.1992.tb04011.x>
 - [16] Yamada, T., Watanabe, S.: On the uniqueness of solutions of stochastic differential equations. *Journal of Mathematics of Kyoto University* **11**(1), 155–167 (1971) <https://doi.org/10.1215/kjm/1250523691>
 - [17] Mishura, Y., Pilipenko, A., Ralchenko, K.: Gatheral double stochastic volatility model with Skorokhod reflection. *Theory of Probability and Mathematical Statistics* **113**, 153–171 (2025) <https://doi.org/10.1090/tpms/1247> arXiv:2505.09184 [q-fin.MF]
 - [18] Andersen, L.B.G., Piterbarg, V.V.: Moment explosions in stochastic volatility models. *Finance and Stochastics* **11**(1), 29–50 (2007) <https://doi.org/10.1007/s00780-006-0011-7>
 - [19] Karatzas, I., Shreve, S.E.: *Brownian Motion and Stochastic Calculus*, 2nd edn. Graduate Texts in Mathematics, vol. 113. Springer, New York, NY (1991). <https://doi.org/10.1007/978-1-4612-0949-2>